PA 541: Advanced Data Analysis I (4 hours)

Spring 2026

Instructor: Michael D. Siciliano
office: 2116 AEH
ph: 312-413-5177
email: sicilian@uic.edu

Class modality, time, and location: This is an in-person class on Mondays, 6:00 pm – 8:30, AEH 2201

Office Hours: Happy to meet at any time. Please email me to set up a convenient time.

Teaching Assistant: Seohyeon (Sunny) Ko (sko78@uic.edu).

Seohyeon Office Hours: Thursdays 12pm to 1pm via Zoom

If these hours don't work, email Seohyeon, she is happy to meet at other times.

I. LEARNING AND COURSE OBJECTIVES:

This course introduces modern econometric and statistical techniques useful for applied researchers and analysts. The primary focus will be on your ability to develop and interpret statistical models effectively, grounded in a strong conceptual understanding of modeling assumptions and limitations. Less emphasis will be placed on mathematical proofs and derivations. Topics covered in this course include standard OLS regression, regression with qualitative predictors, proper handling of data issues, limited dependent variable models, and panel data methods. Directed acyclic graphs (DAGs) will be used to think carefully about data-generating processes and when we can make causal inferences from observational data.

Prerequisites: You should have completed PA 402 or an equivalent course on basic statistics and programming.

By the end of the course, you will be able to:

- Import, clean, and wrangle data in R
- Visualize and summarize data and model results in R
- Use Quarto to produce documents that integrate code, output, and text
- Understand the assumptions of regression models and what to do when those assumptions are violated
- Interpret the output of statistical models
- Test hypotheses about model parameters and exclusion restrictions
- Use qualitative and categorical predictors in statistical models

- Use and interpret interaction terms to assess moderation
- Understand the basics of generalized linear models
- Comprehend the advantages and limitations of panel data; run and interpret panel data models
- Use directed acyclic graphs (DAGs) to model the data-generating process and identify confounders, colliders, and mediators

II. SOFTWARE REQUIREMENTS

We will use the R programming language. R provides broad coverage and availability of new, cutting-edge statistical applications (no need to wait for a new release of the software), as well as in other methodological areas that may be of interest to researchers (e.g., text analysis, spatial analysis, network analysis, QCA, etc.). Learning R will also facilitate your understanding of the literature in your field, as many scholars report their results and provide replication files in R. An active and engaged group of R users provides quality support available through listservs, Stack Exchange, and other resources. R allows you to easily write your own functions to perform tasks unique to your data that would otherwise be quite time-intensive. Finally, and importantly, R is free.

The R code used to produce the data analysis examples and graphics in my lectures will be made available to you. This allows you to read through the lecture notes while simultaneously working in R to reproduce all of the outcomes. Over the years of teaching this course, this has proven to be a useful way to learn data analysis techniques.

III. TEXTS

There is one required text for this class. The second book is optional.

1. Wooldridge, J. M. *Introductory econometrics: a modern approach*: Cengage Learning. 7th edition. [***Required; Earlier editions are fine***]
2. Long, J.D. & Teetor, Paul (2019). *R cookbook*: O'Reilly [***Optional; for those new to R***]

I will not be assigning any specific readings out of the Long and Teetor text. This book can serve as a reference guide for those who are new to R. I will also share a folder through Canvas with a number of free, online R resources.

IV. REQUIREMENTS AND GRADES

1. Midterm and Final

A midterm and final exam will be given. You will be allowed to use a single page (double-sided) of hand-written notes for each exam.

2. Problem sets

Most weeks, you will have a problem set to work on. I usually give problem sets that span two weeks of classes, and I expect to assign around 4 problem sets throughout the semester (depending on our progress and student needs). The problem sets will focus on applying the methods to real data. I encourage you to work with your classmates on these assignments but to ultimately complete and write up the results individually. Please see UIC's policy on academic integrity and this course's policy on the usage of generative AI.

3. Data Analysis Paper

The final component of this course is a relatively short data analysis paper (3,000 – 5,000 words). For this paper, you will use existing datasets from online resources or repositories such as ICPSR, Harvard Dataverse, World Values Survey, IES, Census Data, Pew Internet and American Life Project, Current Population Survey, Open Data Portals (e.g., https://data.cityofchicago.org/) , etc.…or from a researcher willing to share their data. You will be required to develop testable hypotheses using methods covered in this course and to situate your hypotheses within the current literature regarding your topic of interest. The aim of this paper is to give you a more complete experience of the data analysis process. You will go from initial data screening and cleaning to final model output and interpretation. For this assignment, you will write your paper as if preparing it for submission to a journal or as a policy report for a government agency or think tank. In an appendix, you will provide more technical details regarding how you cleaned the data, tested model assumptions, etc. Additional details on the paper will be discussed in class. You should begin thinking about this paper and searching for applicable datasets early on in the semester. You will work in groups of three or four on this paper. Groups should be determined by the third week.

Your grade in this course will be calculated as follows:

Component	Percentage of overall grade	Due date
Midterm	25%	March 16
Final	25%	April 27
Problem Sets (4)	20%	Ongoing
Data Analysis Paper	20%	May 6th
Class Participation	10%	Ongoing

V. WEEKLY SCHEDULE (readings subject to change)

**SCHEDULE OF CLASSES AND ASSIGNMENTS IS SUBJECT TO CHANGE. IN THE EVENT OF ANY CHANGE IN ASSIGNMENT, TOPIC, OR DUE DATE, I WILL POST NOTICE IN THE ANNOUNCEMENTS ON CANVAS AND UPLOAD A CORRECTED SYLLABUS.*

Session 1 - January 12

Topics: Course intro and review. Introduction to Quarto.

*Please download the most recent versions of R and RStudio onto your laptops before class.

R Readings:
Wickham, H. et al. *R for Data Science 2nd ed.*: Chapter 3: Data Transformation: https://r4ds.hadley.nz/

Wickham, H. et al. *R for Data Science 2nd ed.*: Chapter 28: Quarto: https://r4ds.hadley.nz/

Optional R Tutorials (for those who want an R refresher):

Posit Primers: https://posit.cloud/learn/primers/1

January 19 – No Class – MLK Day

Session 2 – January 26

Topics: General discussion of statistical models. The nature of econometric data. Simple regression and simple regression assumptions.

Readings:

Wooldridge Chapters 1 and 2

Session 3 – February 2

Topics: Multiple regression - mechanics and interpretation. Introduction to the *modelsummary* package.

Readings:

Wooldridge Chapter 3

Session 4 – February 9

Topics: Testing hypotheses about model parameters; testing exclusion restrictions.

Readings:

Wooldridge Chapter 4

Session 5 - February 16

Topics: Using qualitative/categorical predictors. Interpreting main effects and interactions with categorical predictors.

Readings:

Wooldridge Chapter 7

Session 6 – February 23

Topics: Non-linear relationships. Checking the validity of regressions assumptions. Dealing with heteroskedasiticty. Issues with highly correlated predictors. Introduction to the *marginaleffects* package.

Readings:

Skim Wooldridge Chapter 8

Skim the *marginaleffects* package sections I and II: Marginal effects

Session 7 – March 2

Topics: Model specification and data issues. Log Models. Data screening and cleaning. Methods for handling outliers and missing data.

Readings:

Wooldridge Chapter 9

Pardoe, I. (2012) Applied Regression Modeling, Chapter 5

Session 8 - March 9

Topics: Limited dependent variables and the generalized linear model. Logistic regression.

Readings:

Agresti, Alan. Chapter 15 - Logistic Regression: Modeling Categorical Responses. P. 687-703.

Optional: Moore, David S. and McCabe, George P., "Introduction to the Practice of Statistics", Chapter 16 – Logistic Regression. P. 1-16

March 16

****Midterm**

March 23 – No Class - UIC Spring Break

Session 9 - March 30

Topics: Panel data analysis - Part I. Pooled cross-sectional analysis and basic difference in difference models.

Readings:

Wooldridge Chapter 13

Session 10 – April 6

Topics: Panel data analysis - Part II. Fixed effects models.

Readings:

Wooldridge Chapter 14

Dougherty, Christopher. (2011) "Introduction to Econometrics", 4th ed. Oxford University Press: New York. Chapter 14 - Introduction to Panel Methods.

Session 11 - April 13

Topics: Introduction to Directed Acyclic Graphs

Readings:

Rohrer, J. M. (2018). Thinking Clearly About Correlations and Causation: Graphical Causal Models for Observational Data. *Advances in Methods and Practices in Psychological Science, 1*(1), 27-42. doi:10.1177/2515245917745629

Shrier, I., & Platt, R. W. (2008). Reducing bias through directed acyclic graphs. *BMC Medical Research Methodology, 8*(1), 70. doi:10.1186/1471-2288-8-70

OPTIONAL: Here are some useful links and tutorials: http://dagitty.net/learn/

Session 12 - April 20

Topics: Causal Inference and the Table 2 Fallacy. Catch up and Review for Final Exam.

Readings:

Westreich, D., & Greenland, S. (2013). The Table 2 Fallacy: Presenting and Interpreting Confounder and Modifier Coefficients. *American Journal of Epidemiology*, 177(4), 292–298.

Hünermund, P., & Louw, B. (2023). On the Nuisance of Control Variables in Causal Regression Analysis. *Organizational Research Methods*, 10944281231219274.

April 27

****In Class Final Exam**

FINALS WEEK

****Data Analysis Paper Due on May 6th**

VI. OTHER RULES AND RESOURCES

Policy for Missed or Late Work: Assignments and papers are expected to be submitted on time. The schedule is also designed to help you manage your workload and it is hard to keep up with the class material if you fall behind on assignments and readings. That said, life happens. Late work may be accepted if you communicate with me and the TAs. Please ask for an extension before the due date and indicate when you plan to submit the work.

Attendance and Participation Policy: Please email me and the TAs if you face an unexpected situation that may impede your attendance or timely completion of assignments. Attendance each week and active participation in class discussions and exercises is expected.

Community Agreement/Classroom Conduct Policy: Data analysis is often a team sport. As such, it is important that you are respectful of others in the classroom. Please be sure to turn off and silence cell phones. Cell phones are not to be out during the class. Studies have shown your cognitive capacity is reduced by just having your smartphone sitting out next to you. Having and using your phone not only lowers your ability to learn and your likelihood of success in a class, but it also harms the learning of those around you. As a member of this classroom, I also expect all of us to engage in professional dialogue; debate concepts and ideas but not individuals; and be willing to work together to help one another and share study strategies, coding tips, and other useful information.

Academic Integrity: As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, and administrators—share the responsibility of ensuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students shall be handled pursuant to the Student Disciplinary Policy. The

Student Disciplinary Policy is available online at https://dos.uic.edu/wp-content/uploads/sites/262/2018/10/DOS-Student-Disciplinary-Policy-2018-2019-FINAL.pdf.

Use of ChatGPT and other Generative Artificial Intelligence: The use of generative AI is acceptable in certain situations in this class. For instance, you can use ChatGPT to help troubleshoot code or obtain working examples of code to understand how a specific function operates. In other situations, you are prohibited from using generative AI. For instance, using features like ChatGPT's advanced data analysis tools, which allow you to upload datasets and ask ChatCPT to answer questions on the problem sets, is strictly prohibited. Further, you cannot use generative AI to write the interpretation of your analyses or sections of your final paper. We will discuss appropriate uses and ways to integrate ChatGPT into your workflow as the semester progresses. If you have any questions as to whether and when generative AI tools can be used in this course, please reach out.

Special Needs: UIC and the Department of Public Administration are committed to maintaining a barrier-free environment so individuals with disabilities can fully access programs, services and all activities on campus. The Office of Disability Services works to ensure the accessibility of UIC programs, classes, and services to students with disabilities. Services are available for students who have documented disabilities, including vision or hearing impairments and emotional or physical disabilities. Students with disability/access needs or questions may contact the Office of Disability Services at (312) 413-2183 (voice) or (312) 413-0123 (TTY only). Please feel free to contact me if you need any special accommodations.

Diversity and Inclusion: It is my goal that people from diverse backgrounds and perspectives be included in and served by this course, the Department of Public Administration, and the University of Illinois at Chicago. I believe that the diversity that the students bring to this class is a resource that can be used to improve student learning and perspectives. Course materials and activities are designed to be respectful of all types of diversity: gender identity, sexuality, disability, age, socioeconomic status, ethnicity, race, nationality, religion, values, and culture. It is my intent that this class be an environment where all students feel safe to express their perspectives and opinions. Please let me know if something said or done by me, the TA, guest lecturers, or other students, is troubling or causes discomfort or offense. If this occurs, please feel free to discuss the situation with me privately, to raise your concern in class, or to let me know about the issue through a trusted sources (e.g., another student or faculty member or your academic advisor).

Campus Advocacy Network: Under the Title IX law you have the right to an education that is free from any form of gender-based violence and discrimination. Crimes of sexual assault, domestic violence, sexual harassment, and stalking are against the law and can be prevented. For more information or for confidential victim-services and advocacy contact UIC's Campus Advocacy Network at 312-413-1025 or visit http://can.uic.edu/. To make a report to UIC's Title IX office, contact Rebecca Gordon, EdD at TitleIX@uic.edu or (312) 996-5657.